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Bridging 2D and 3D Segmentation Networks for Computation Efficient Volumetric Medical Image Segmentation: An Empirical Study of 2.5D Solutions

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Recently, deep convolutional neural networks have achieved great success for medical image segmentation. However, unlike segmentation of natural images, most medical images such as MRI and CT are volumetric data. In order to make full use of volumetric information, 3D CNNs are widely used. However, 3D CNNs suffer from higher inference time and computation cost, which hinders their further clinical applications. Additionally, with the increased number of parameters, the risk of overfitting is higher, especially for medical images where data and annotations are expensive to acquire. To issue this problem, many 2.5D segmentation methods have been proposed to make use of volumetric spatial information with less computation cost. Despite these works lead to improvements on a variety of segmentation tasks, to the best of our knowledge, there has not previously been a large-scale empirical comparison of these methods. In this paper, we aim to present a review of the latest developments of 2.5D methods for volumetric medical image segmentation. Additionally, to compare the performance and effectiveness of these methods, we provide an empirical study of these methods on three representative segmentation tasks involving different modalities and targets. Our experimental results highlight that 3D CNNs may not always be the best choice. Despite all these 2.5D methods can bring performance gains to 2D baseline, not all the methods hold the benefits on different datasets. We hope the results and conclusions of our study will prove useful for the community on exploring and developing efficient volumetric medical image segmentation methods.

2D-2.5D-3D

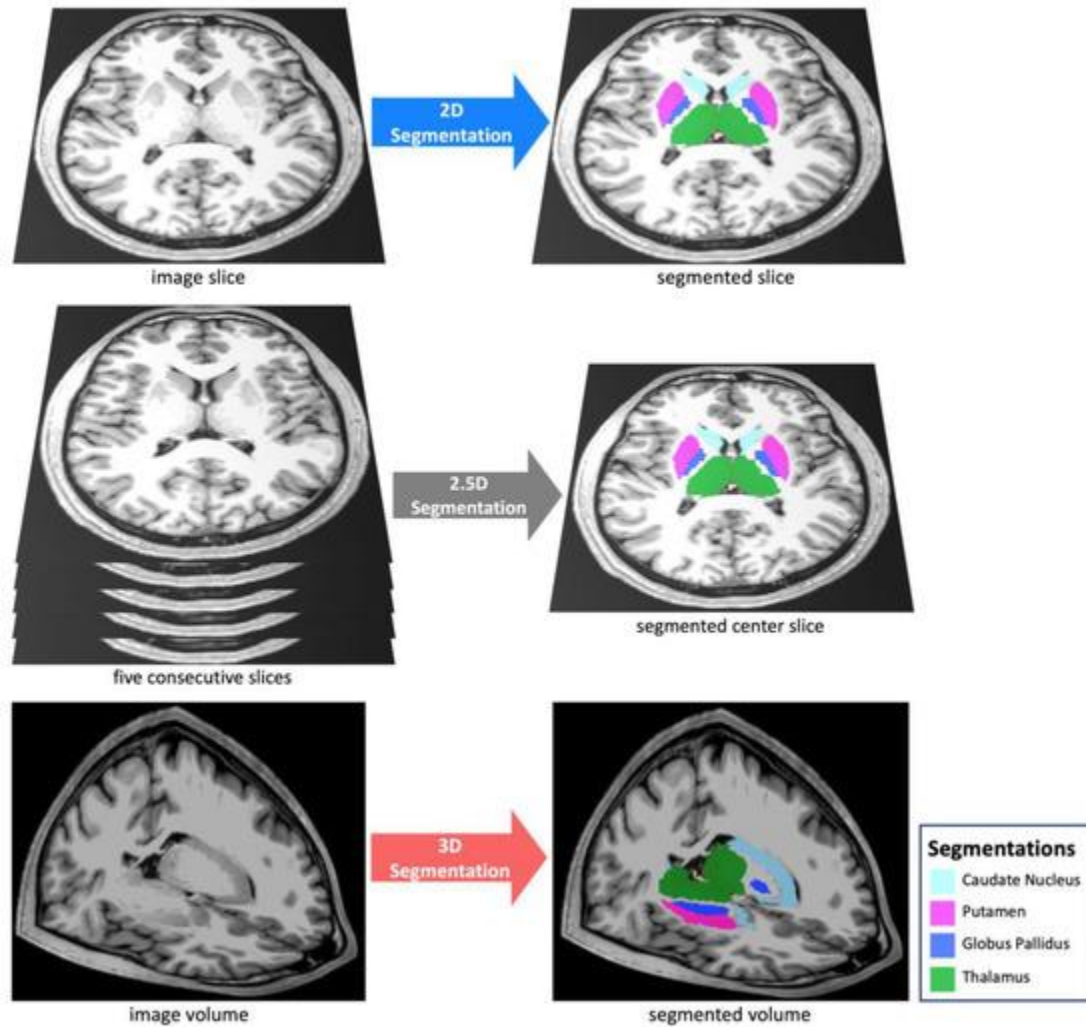
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Introduction

Background



Related Work

Multi-view Fusion

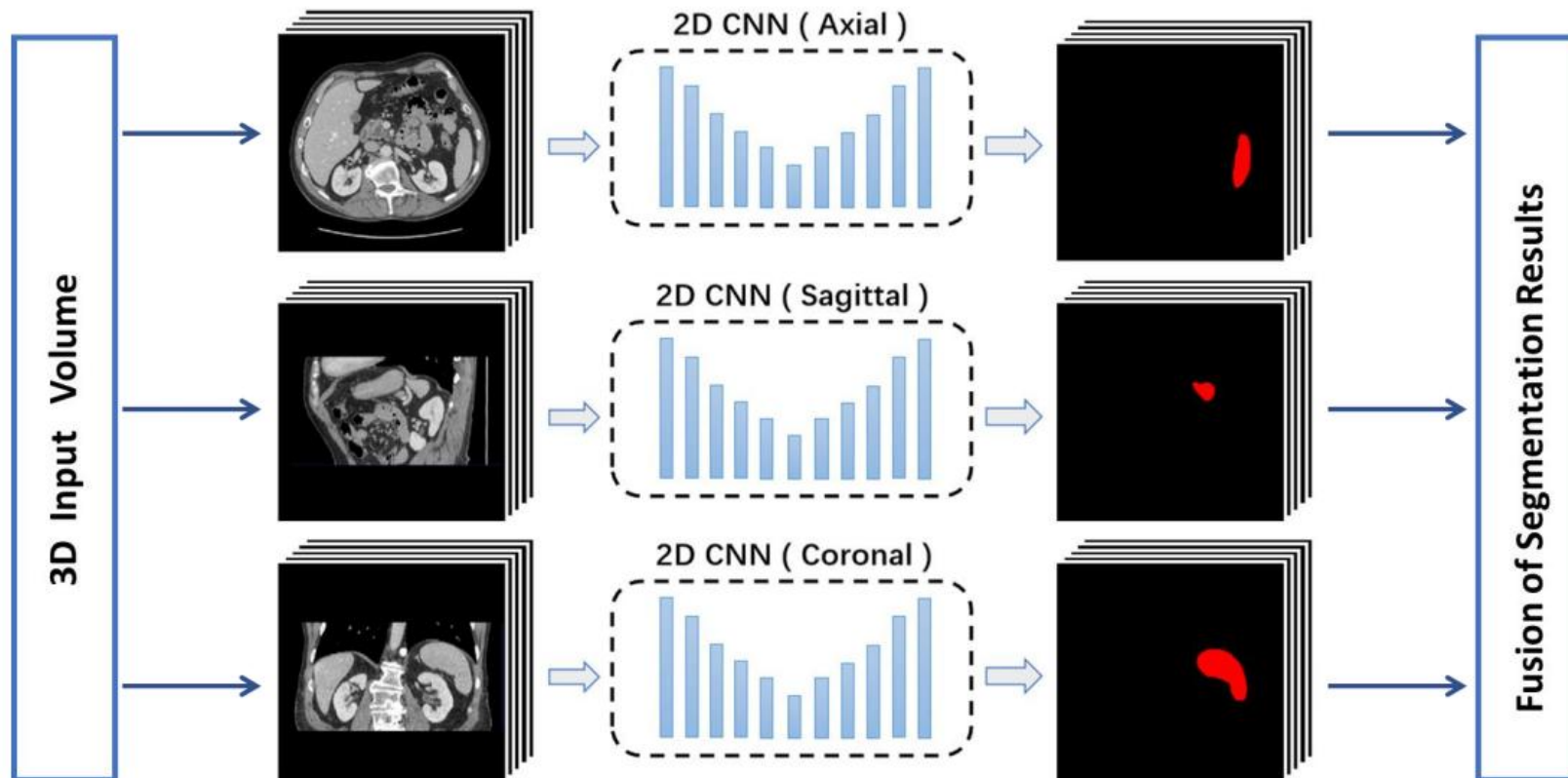
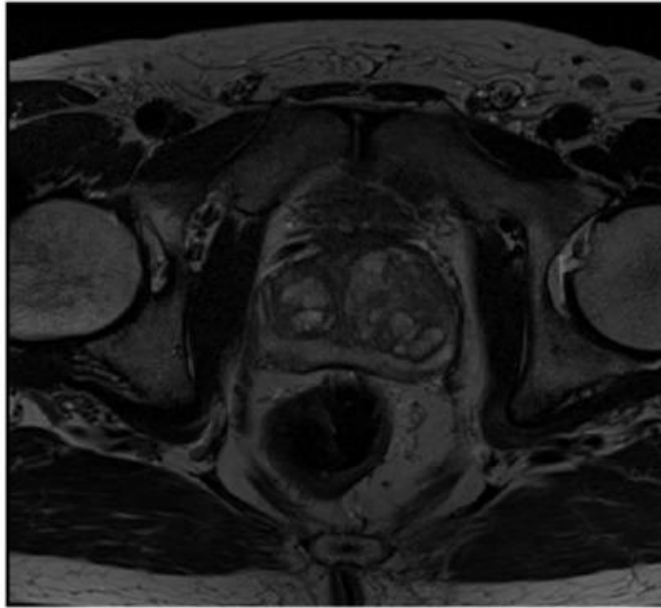


Figure 2: The overall workflow of multi-view fusion. The 3D input volume is divided into 2D slices from axial, sagittal and coronal planes to train the corresponding 2D CNNs. Then the segmentation results from different views are ensembled to get the final 2.5D segmentation result.

Related Work

Multi-view Fusion



Short-axis Slices of Prostate MRI



Long-axis Slices of Prostate MRI

Figure 3: Examples of short-axis slices and long axis slices for anisotropic volumetric medical images. From the figure we can see that long-axis slices are with poor contextual information due to the low resolution on z axis.

Related Work

Inter-slice Information

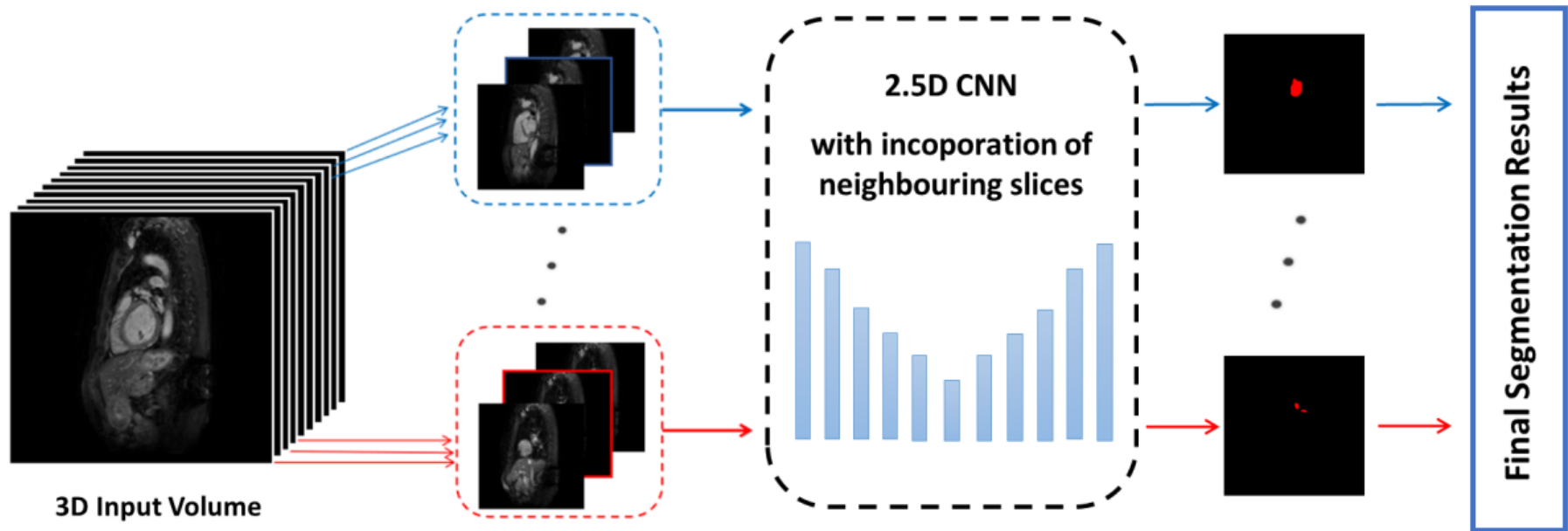


Figure 4: The overall workflow of incorporating inter-slice information. With the additional input of neighboring slices, the inter-slice correlation could be leveraged so as to get better performance compared with 2D CNNs.

Related Work

Fusing 2D/3D Features

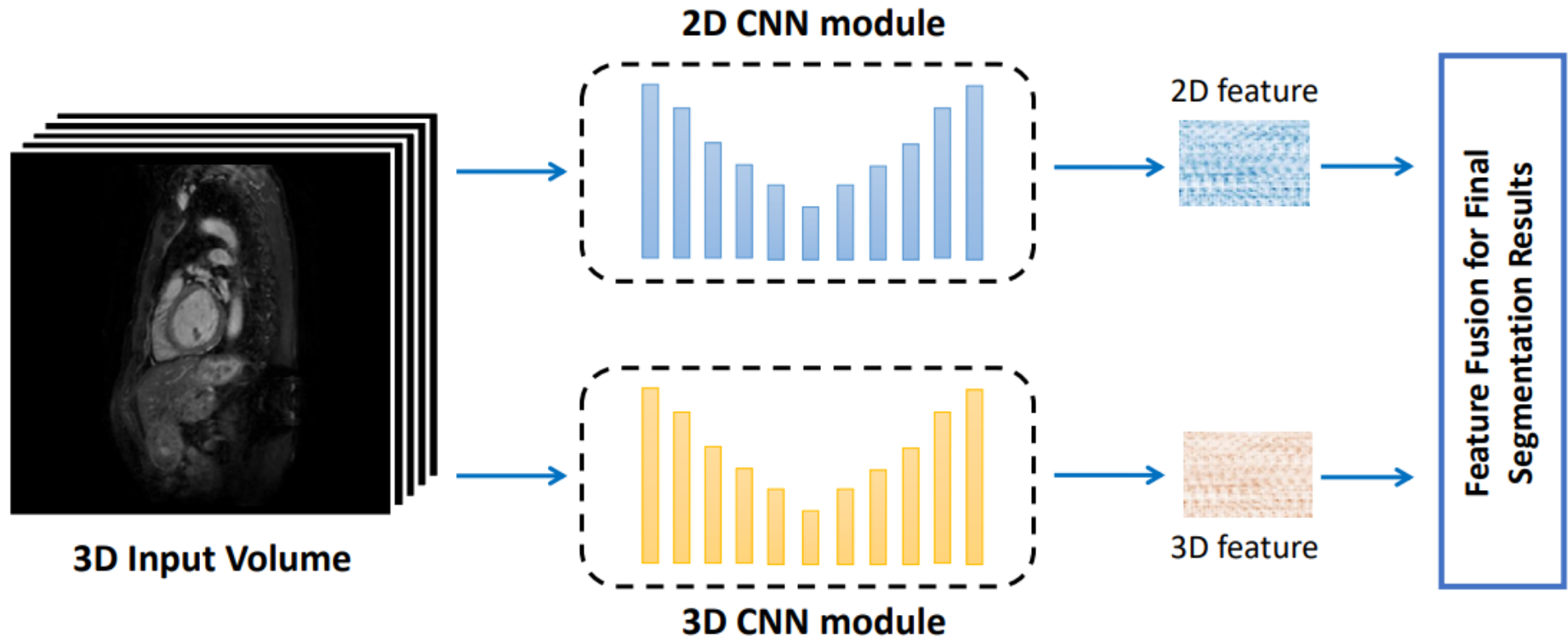
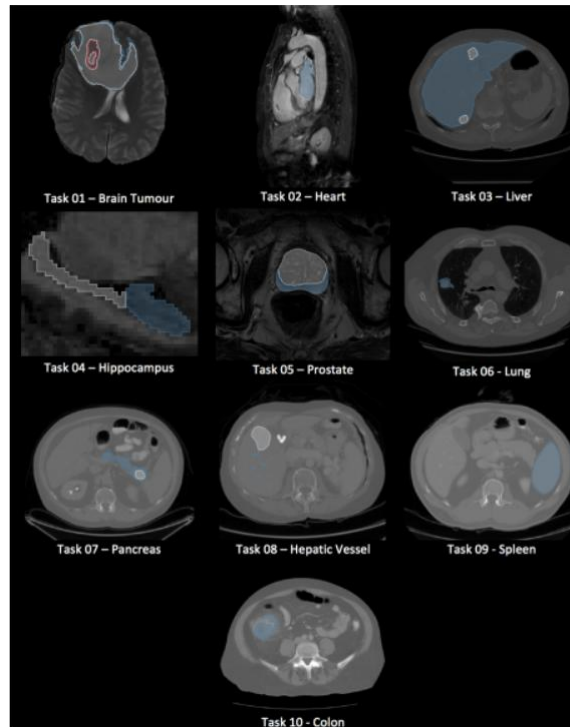


Figure 5: The overall workflow of fusing 2D/3D features. Features extracted from 2D and 3D CNNs/modules are fused to utilize volumetric spatial information.

Experiments

Datasets



Dataset	Modality	Volume Count	Source Institution	Target
Cardiac	MRI (mono)	20	King's College London	Left Atrium
Spleen	CT	41	Memorial Sloan Kettering Cancer Center	Spleen
Prostate	MRI (T2, ADC)	32	Radboud University	Prostate

Experiments

Backbone structure and evaluation metrics

Item	Description
Backbone	U-Net with 4-stage convolutional blocks (base: 8 feature maps)
Comparison Model	3D U-Net with identical environment and settings
Optimizer	Adam
Initial LR	0.001 (reduced by factor of 0.5 if no improvement in 20 epochs)
Loss Function	Cross-Entropy Loss + Dice Loss
Evaluation Metrics	Dice Similarity Coefficient (DSC), 95% Hausdorff Distance (95HD)

Experiments

results

Table 1. Quantitative segmentation results of 2D U-Net from different planes, 2.5D methods with different fusion strategies and 3D U-Net on the evaluation datasets. The up arrow indicates that larger value of metric is better and vice versa. Bolded results indicate best segmentation plane in 2D CNNs and best fusion methods in 2.5D CNNs. Underlined results indicate 2.5D results that have better performance compared with 3D CNNs.

Dataset	Metrics	2D CNNs			2.5D Multi-view Fusion			3D CNNs
		U-Net (A)	U-Net (S)	U-Net (C)	MV	WV	VFN	3D U-Net
Cardiac MRI	<i>Dice</i> ↑	0.8362	0.8630	0.8225	0.8992	0.8996	0.9140	0.9203
	<i>95HD</i> ↓	7.120	3.274	5.268	1.287	1.104	1.021	1.014
Spleen CT	<i>Dice</i> ↑	0.9114	0.8027	0.8858	0.9204	0.9208	0.9410	0.9452
	<i>95HD</i> ↓	1.521	8.710	2.165	1.511	1.508	<u>1.119</u>	1.222
Prostate MRI	<i>Dice</i> ↑	0.7222	0.4322	0.4545	0.6448	0.7490	<u>0.7731</u>	0.7677
	<i>95HD</i> ↓	8.933	23.72	15.31	4.956	4.402	<u>4.460</u>	5.935

Part 3, Experiments

results

Table 2. Quantitative segmentation results of 2D and 2.5D methods with different number of input channels on the evaluation datasets. The up arrow indicates that larger value of metric is better and vice versa. Bolded results indicate best number of slices for multi-slice input. Underlined results indicate 2.5D results that have better performance compared with 3D CNNs.

Dataset	Metrics	2D CNNs	2.5D Multi-slice Input			3D CNNs
		U-Net (1slice)	3 slices	5 slices	7 slices	3D U-Net
Cardiac MRI	<i>Dice</i> ↑	0.8630	0.8781	0.8642	0.867	0.9203
	<i>95HD</i> ↓	3.274	1.559	2.525	3.837	1.014
Spleen CT	<i>Dice</i> ↑	0.9114	0.9121	0.9082	0.9089	0.9452
	<i>95HD</i> ↓	1.521	1.345	1.581	1.527	1.222
Prostate MRI	<i>Dice</i> ↑	0.7222	0.7460	0.7448	0.7257	0.7677
	<i>95HD</i> ↓	8.933	5.565	<u>4.759</u>	7.225	5.935

Part 3, Experiments

results

Table 3. Quantitative segmentation results of 2.5D methods to incorporate inter-slice information, 2D U-Net and 3D U-Net. The up arrow indicates that larger value of metric is better and vice versa. Bolded results indicate best methods to incorporate inter-slice information. Underlined results indicate 2.5D results that have better performance compared with 3D CNNs.

Dataset	Metrics	2D CNNs	2.5D Incorporate inter-slice information				3D CNNs
		U-Net	Multi-slice	RNN	Att Shape	Att Border	3D U-Net
Cardiac MRI	<i>Dice</i> ↑	0.8630	0.8781	0.8858	0.8821	0.8747	0.9203
	<i>95HD</i> ↓	3.274	1.559	2.155	1.494	1.750	1.014
Spleen CT	<i>Dice</i> ↑	0.9114	0.9121	0.9167	0.9233	0.9155	0.9452
	<i>95HD</i> ↓	1.521	1.345	1.573	1.390	1.794	1.222
Prostate MRI	<i>Dice</i> ↑	0.7222	0.7460	0.7801	0.7804	<u>0.7885</u>	0.7677
	<i>95HD</i> ↓	8.933	5.565	4.914	<u>3.448</u>	3.907	5.935

Part 3, Experiments

results

Table 4. Quantitative segmentation results of different stages and methods to fuse 2D and 3D features. The up arrow indicates that larger value of metric is better and vice versa. Bolded results indicate best fusion methods in 2.5D CNNs. Underlined results indicate 2.5D results that have better performance compared with 3D CNNs.

Dataset	Metrics	2D CNNs	Fusion of 2D and 3D features				3D CNNs
			Encoder	Encoder	Output	Output	
			Add	SE	Add	SE	
Cardiac MRI	<i>Dice</i> ↑	0.8630	0.8743	0.8732	0.8876	0.8824	0.9203
	<i>95HD</i> ↓	3.274	1.759	1.913	1.604	1.707	1.014
Spleen CT	<i>Dice</i> ↑	0.9114	0.9221	0.9193	0.9197	0.9122	0.9452
	<i>95HD</i> ↓	1.521	1.445	1.463	1.425	1.487	1.222
Prostate MRI	<i>Dice</i> ↑	0.7222	0.7562	<u>0.7693</u>	0.7625	0.7597	0.7677
	<i>95HD</i> ↓	8.933	7.141	6.259	<u>5.567</u>	8.780	5.935